Latest version: https://vuphan314.github.io
This version: 2023 Mar 06

Vu H. N. Phan

Curriculum Vitae

Portland, Oregon
vuphan314.github.io
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Vu Phan is a Formal Verification engineer at Intel, where he performs pre-silicon validation of a hardware design. Vu earned a Ph.D. degree in Computer Science at Rice University; his advisor was Prof. Moshe Vardi.

Experience

2022-now **Computer Engineer**, *Intel Corporation*
- Perform pre-silicon Formal Verification of an Intellectual Property in a System on a Chip

2021 **Technical Intern**, *Intel Corporation*
- Developed software to benchmark Formal Verification platforms

Certificates

2023 Mar **Jasper Formal Fundamentals v21.09**, *Cadence Design Systems*

2021 Jul **SVA, Formal, and JasperGold Fundamentals for Designers v19.03**, *Cadence Design Systems*

Languages

Hardware **SystemVerilog**

Software **Python, C++**

Education

2020-2022 **Doctor of Philosophy in Computer Science**, *Rice University*, GPA 3.63/4.00

2017-2019 **Master of Science in Computer Science**, *Rice University*, GPA 3.60/4.00

2014-2017 **Bachelor of Science in Computer Science & Math**, *Texas Tech University*, GPA 4.00/4.00

2013-2014 **Associate of Science**, *Collin College*, GPA 4.00/4.00

Projects

2019-2021 **DPMC**, *Dynamic-Programming Model Counter*, https://github.com/vardigroup/DPMC
- Implemented a weighted #SAT framework with Jeffrey Dudek

2018-2019 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, https://github.com/vardigroup/ADDMC
- Implemented a weighted #SAT solver

2016-2018 **LED**, *Language of Effective Definitions*, https://vuphan314.github.io/LED
- Translated the literate-programming language LED into SequenceL and LaTeX

2015-2016 **L**, *Logic*, https://github.com/iensen/LtoASPtranslator
- Translated the logic-programming language L into Answer Set Prolog with Evgenii Balai

Courses

2017-2021 **Graduate Level**

- Reasoning about Software (languages: Dafny, Coq)
- Bioinformatics: Sequence Analysis
- Statistical Machine Learning
- Computer Systems Architecture
- Multi-Core Computing
- Compiler Construction
- Artificial Intelligence
- Programming Languages
- Logic in Computer Science

2014-2017 **Undergraduate Level**

- Operating Systems
- Database Systems
- Computer Architecture
- Programming Languages
- Software Engineering
- Algorithms
- Object-Oriented Programming
- Computer Organization and Assembly Language Programming
- Modern Digital System Design
- Data Structures
- Automata
- Programming Principles

Bibliography

Vu Phan is the correspondence author of the following publications (authors are sorted by surnames).

Theses

[Pha22] Vu H. N. Phan. "Quantitative Reasoning on Hybrid Formulas with Dynamic Programming". PhD thesis. Rice University, 2022. URL: https://scholarship.rice.edu/handle/1911/113243.

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: https://scholarship.rice.edu/handle/1911/107761.

Conference Papers

[DPV21] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ProCount: Weighted Projected Model Counting with Graded Project-Join Trees". In: *Conference on Theory and Applications of Satisfiability Testing (SAT)*. 2021. URL: http://jeffreydudek.com/papers/sat21_procount.pdf.

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: https://arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: https://arxiv.org/abs/1907.05000.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: https://arxiv.org/abs/1809.09319.